Matrices (AI HL)

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

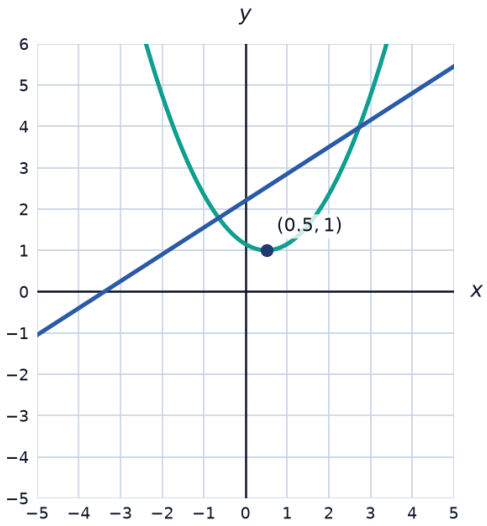
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The temperature θ of a cooling metal rod satisfies $\frac{d\theta}{dt} = -k(\theta - 20)$, where t is time in minutes and k is a constant. Given $\theta = 80$ when $t = 0$ and $\theta = 50$ when $t = 10$, find the value of t when $\theta = 30$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the constant term (independent of x) in the binomial expansion of $\left(2x^2 - \frac{1}{x}\right)^9$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The curve C has equation $x^3 - 2xy + y^2 = 4$. Find the coordinates of the two points on C where the tangent is parallel to the y-axis (vertical).

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A bag contains 5 red and 4 green marbles. Three marbles are drawn at random without replacement. Find the probability that: (a) all three are red, (b) at least two are red.

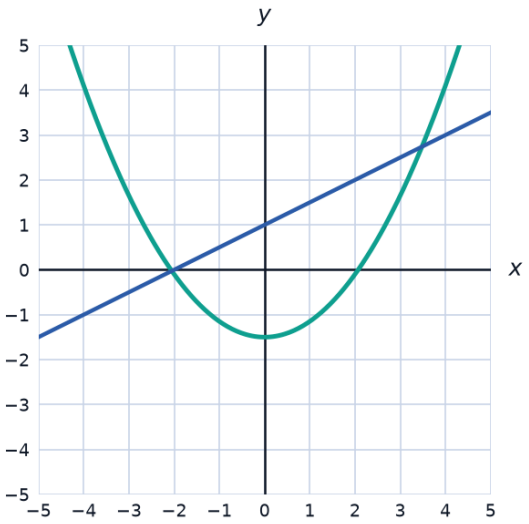
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A train accelerates uniformly from rest to 30 m/s in 20 s, travels at constant speed 30 m/s for 60 s, and then decelerates uniformly to rest in 15 s. Find: (a) the acceleration in the first phase, (b) the total distance travelled, (c) the average speed for the entire journey.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A block of mass $4\ \text{kg}$ is pulled along a rough horizontal plane ($\mu = 0.25$) by a pulling force of magnitude $25\ \text{N}$ acting at 30° above the horizontal. Taking $g = 9.8\ \text{m/s}^2$: (a) show that the normal reaction $R = 26.7\ \text{N}$, (b) find the acceleration of the block.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two particles P of mass $3\ \text{kg}$ and Q of mass $5\ \text{kg}$ are connected by a light inextensible string passing over a smooth fixed pulley. The system is held at rest and released. Taking $g = 9.8\ \text{m/s}^2$: (a) find the acceleration a of the system, (b) calculate the tension T in the string, (c) find the force exerted by the string on the pulley.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A continuous random variable X has PDF given by $f(x) = \frac{k}{x^2}$ for $1 \le x \le 4$, and 0 elsewhere. (a) Show that $k = \frac{4}{3}$. (b) Find the median of X.

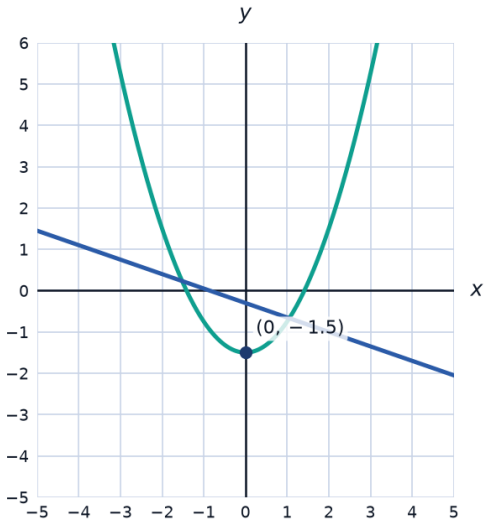
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Write the standard equation of the ellipse having foci at $(\pm 3, 0)$ and vertices at $(\pm 5, 0)$. State the lengths of the major and minor axes, and compute the eccentricity.

© ClipSAT — clipsat.org | All Rights Reserved

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An ellipse has vertices at $(1, 6)$ and $(1, -4)$, and a focus at $(1, 4)$. Determine its standard equation and compute its eccentricity.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A car of mass $1200\ \text{kg}$ ascends a hill of inclination α, where $\sin\alpha = \frac{1}{20}$, against a constant resistance to motion of $500\ \text{N}$. Taking $g = 9.8\ \text{m/s}^2$: (a) find the driving force required to travel at a steady speed of $25\ \text{m/s}$, (b) calculate the power developed by the engine in kilowatts.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A radioactive substance follows the decay model $M(t) = M_0 e^{-kt}$. If 20% of the sample decays in 15 days, find the exact decay constant k and the substance's half-life in days to the nearest tenth.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

On an Argand diagram, the point P represents the complex number z such that $|z - (2 + 3i)| \leq 2$. Find the greatest possible value of $|z|$ for points in this region.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Events A and B are such that $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cup B) = 0.8$. Find $P(A \cap B)$, calculate $P(A|B)$, and determine whether A and B are independent.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

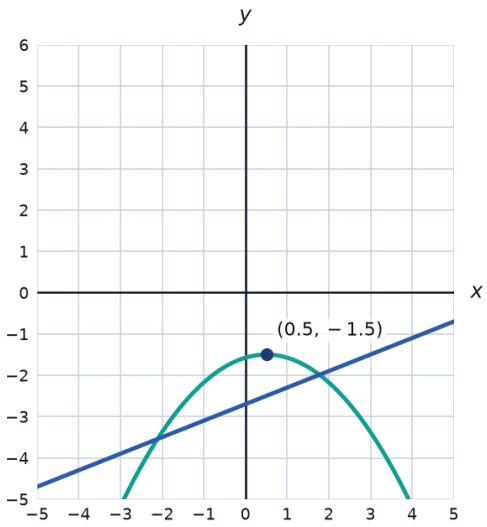
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.02 Differential Equations

Find the equation of the perpendicular bisector of the line segment joining $A(-2, 5)$ and $B(4, -3)$. Give your answer in the form $ax + by + c = 0$, where a, b, c are integers.

© ClipSAT — clipsat.org | All Rights Reserved

16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series**

The random variable $X \sim N(\mu, \sigma^2)$ satisfies $P(X < 65) = 0.8413$ and $P(X < 45) = 0.0228$. Find the values of μ and σ.



17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)**

Express $\frac{3+\sqrt{5}}{2-\sqrt{5}}$ in the form $p + q\sqrt{5}$, where p and q are rational numbers to be found.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)**

Level 2 · Medium Solve the initial value problem $\frac{dy}{dx} - y = e^{3x}$ with $y(0) = 2$.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.03 Markov Chains (AI HL)**

Level 1 · FoundationsExpress the infinite summation limit as a definite integral on the interval $[0, 1]$ and evaluate it: $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{i^3}{n^4}$.

20 **[Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Proof by Mathematical Induction**

Solve the logarithmic equation $\log_2(x+3) + \log_2(x-3) = 4$ for all real solutions x. State any extraneous roots.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Compute the matrix product AB for $A = [[1, 2], [3, 4]]$ and $B = [[2, 0], [1, 3]]$.

a. $[[4, 6], [10, 12]]$

b. $[[2, 0], [3, 12]]$

c. $[[4, 10], [6, 12]]$

d. $[[3, 2], [4, 7]]$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the inverse of the matrix $M = [[3, 5], [1, 2]]$.

a. $[[2, -5], [-1, 3]]$

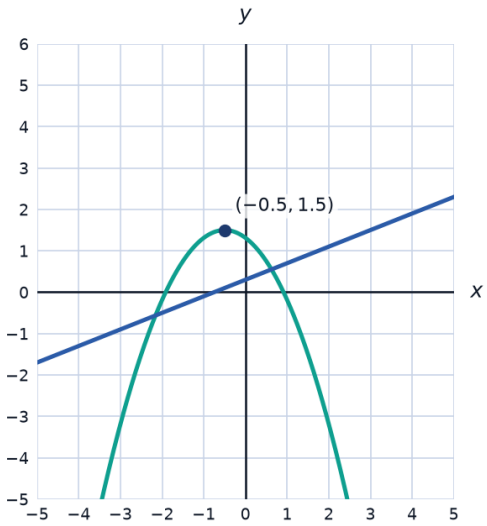
b. $[[-2, 5], [1, -3]]$

c. $[[3, -1], [-5, 2]]$

d. $[[1/3, 1/5], [1, 1/2]]$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If A and B are invertible matrices of the same dimension, what is $(AB)^{-1}$?



a. $B^{-1}A^{-1}$

b. $A^{-1}B^{-1}$

c. BA^{-1}

d. AB^{-1}

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For what value of k is the matrix $A = [[k, 6], [2, 3]]$ singular (non-invertible)?

a. $k = 4$

b. $k = 6$

c. $k = 0$

d. $k = 9$

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If A is a 3×3 matrix with $\det(A) = 5$, what is $\det(2A)$?

a. 40

b. 10

c. 20

d. 80

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the eigenvalues of the matrix $A = [[4, 2], [1, 3]]$.

a. $\lambda = 5$ and $\lambda = 2$

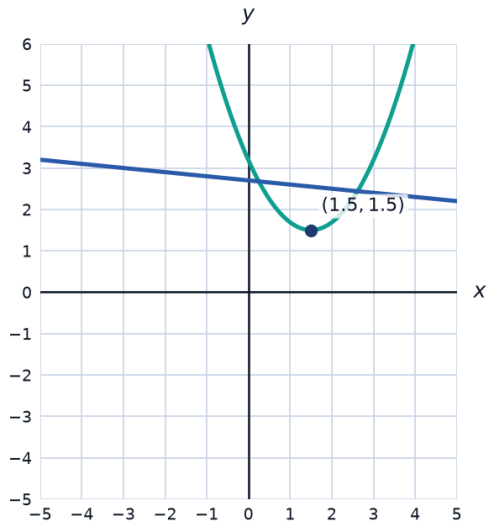
b. $\lambda = 4$ and $\lambda = 3$

c. $\lambda = 6$ and $\lambda = 1$

d. $\lambda = 7$ and $\lambda = 0$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find an eigenvector corresponding to $\lambda = 5$ for $A = [[4, 2], [1, 3]]$.



a. $[2, 1]^T$

b. $[1, 2]^T$

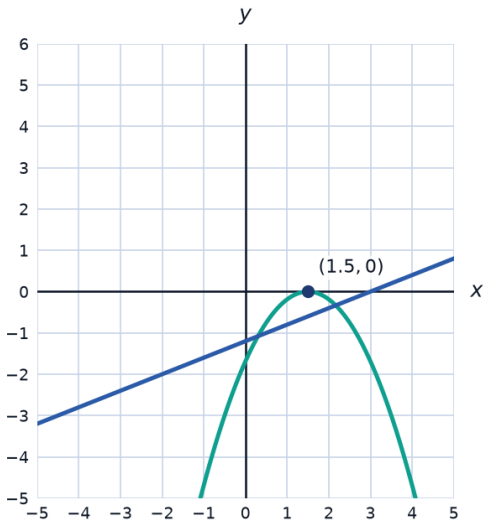
c. $[1, -1]^T$

d. $[2, -1]^T$

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the sum of the eigenvalues of an $n \times n$ matrix A equal to?



a. The trace of A (sum of diagonal entries)

b. The determinant of A

c. The rank of A

d. n

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If a matrix A is diagonalized as $A = PDP^{-1}$, how is the power A^n computed efficiently?

a. $A^n = PD^nP^{-1}$

b. $A^n = P^nD^n(P^{-1})^n$

c. $A^n = D^n$

d. $A^n = PDP^{-1}n$

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

When is an $n \times n$ matrix A guaranteed to be diagonalizable?

a. When it has n linearly independent eigenvectors (or n distinct eigenvalues)

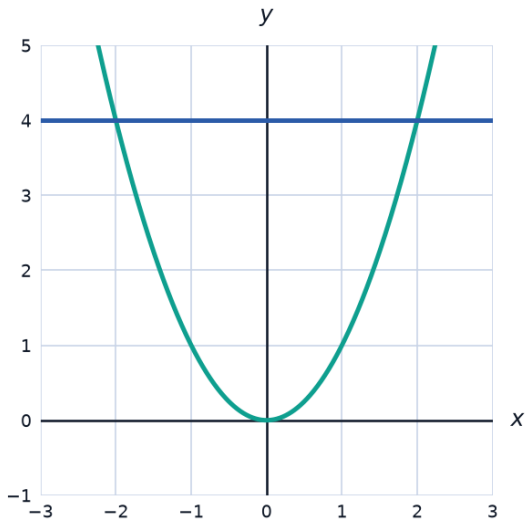
b. When its determinant is non-zero

c. Only if it is a symmetric identity matrix

d. When all its entries are positive

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the region R in the xy-plane bounded by the curves $y = x^2$ and $y = 4$. What is the area of this region?



a. $\frac{32}{3}$

b. $\frac{16}{3}$

c. 8

d. 16

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the solid E bounded by the surfaces $z = x^2 + y^2$ and $z = 8 - x^2 - y^2$. Calculate the volume of this solid using a triple integral by first transforming to cylindrical coordinates. Express your answer in terms of π.

a. $\frac{16\pi}{3}$

b. $\frac{32\pi}{3}$

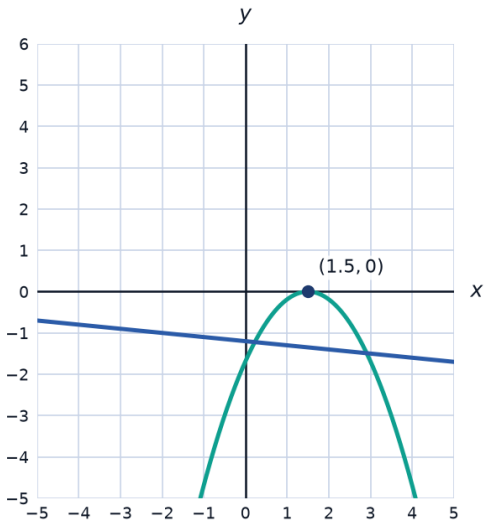
c. 16π

d. $\frac{64\pi}{3}$

© ClipSAT — clipsat.org | All Rights Reserved

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Two lines have slopes $m_1 = -1$, $m_2 = -1$ and y-intercepts $b_1 = -2$, $b_2 = -2$. Classify the system.



a. Consistent and independent (one solution)

b. Consistent and dependent (infinitely many solutions)

c. Inconsistent (no solution)

d. Cannot be determined

34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Probability: $f(x, y) = 6xy$ on $0 \le x \le 1, 0 \le y \le \sqrt{1 - x^2}$ (quarter circle). $P(X + Y \le 1) =$

a. $\frac{3}{4}$

b. $\frac{1}{2}$

c. $\frac{1}{4}$

d. $\frac{2}{3}$

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Evaluate $\int_0^2 \int_y^2 e^{x^2}\, dx\, dy$ by reversing the order of integration.

a. $-\frac{1}{2} + \frac{e^4}{2}$

b. $-1 + e^4$

c. $\frac{1}{2} + \frac{e^4}{2}$

d. $-\frac{1}{4} + \frac{e^4}{4}$

© ClipSAT — clipsat.org | All Rights Reserved

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Use $x = u + v, y = u - v$ to compute $\iint_D (x - y)e^{x+y}\,dA$ where D is the square $|x - y| \le 1, |x + y| \le 1$:

a. 0

b. $2(e - e^{-1})$

c. $(e - 1)^2$

d. $4(e - e^{-1})/e$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Iterated integral over tetrahedron $x + y + z \le 1, x, y, z \ge 0$: volume = $1/6$. The integral $\iiint x\,dV =$

a. $1/6$

b. $1/24$

c. $1/12$

d. $1/8$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Evaluate $\iiint_B (x^2 + y^2 + z^2)\,dV$ where B is the ball $x^2 + y^2 + z^2 \le 1$.

a. $\frac{4\pi}{5}$

b. $\frac{4\pi}{3}$

c. $\frac{2\pi}{5}$

d. $\frac{8\pi}{5}$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.03 Markov Chains (AI HL)

The volume of the region $1 \le x^2 + y^2 + z^2 \le 4$ (shell between spheres of radii 1 and 2):

a. $4\pi/3$

b. 4π

c. 8π

d. $4\pi(8 - 1)/3 = 28\pi/3$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Proof by Mathematical Induction

Use spherical coordinates to find the volume of the first-octant part of the ball $x^2 + y^2 + z^2 \le 16$.

a. $\frac{256\pi}{3}$

b. $\frac{64\pi}{3}$

c. $\frac{32\pi}{3}$

d. $\frac{16\pi}{3}$

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Find the volume of the solid inside the sphere $\rho = 2$ and above the cone $\phi = \frac{\pi}{4}$.

a. $2\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$

b. $4\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$

c. $\frac{16\pi}{3}$

d. $2\sqrt{2}\,\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

The Jacobian of the transformation $x = r\cosh u, y = r\sinh u$ is r times:

a. $\cosh^2 u - \sinh^2 u = 1$

b. $\cosh u + \sinh u$

c. $\sinh^2 u$

d. r^2

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.04 Partial Fractions

Centre of mass of lamina $D = [0,1] \times [0,1]$ with density $\delta = xy$:

a. $(1/2, 1/2)$

b. $(2/3, 2/3)$

c. $(3/4, 3/4)$

d. $(1, 1)$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Use polar coordinates to evaluate $\iint_R (x^2 + y^2)^3\,dA$, where R is the disk $x^2 + y^2 \le 9$.

a. $\frac{6561}{4}$

b. $\frac{6561\pi}{4}$

c. $\frac{6561\pi}{2}$

d. $\frac{6561\pi}{8}$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

The double integral over region D of $f = 1$ gives:

a. area of D

b. perimeter of D

c. volume

d. 0

© ClipSAT — clipsat.org | All Rights Reserved